Big Data - Redis

Author

- Laurent d'Orazio, Univ Rennes, CNRS, IRISA

1 Introduction

The application to be considered relies on a collection of movies. A sample of the data sets is illustrated by figure 1. The attributes of a movie are two identifiers (id and idimdb), the release data, the runtime, the budget and revenues, a link to a poster, an overview and a rating.

idimdb	title	release	runtime	budget	revenue	poster	overview	rating	last_update
tt0029583	Snow White and the Seven Dwarfs	1937-12-21	83	1488423	184925486	https://image.tmdb.org/t/p/original/1P9eGGlT7eV7kA...	A beautiful girl, Snow White, takes refuge in the ...	7.6	2022-02-27
tt0032910	Pinocchio	1940-02-23	88	2600000	84300000	https://image.tmdb.org/t/p/original/ynkEEi296ofs9k...	Lonely toymaker Geppetto has his wishes answered w...	7.4	2022-02-27
tt0033563	Dumbo	1941-10-31	64	812000	1600000	https://image.tmdb.org/t/p/original/hKDdllslMtsU9J...	Dumbo is a baby elephant born with over-sized ears...	7.2	2022-02-27
tt0034492	Bambi	1942-08-14	70	858000	267447150	https://image.tmdb.org/t/p/original/wV9e2y4myJ4KMF...	Bambi's tale unfolds from season to season as the ...	7.3	2022-02-27
tt0042332	Cinderella	1950-02-22	74	2900000	263591415	https://image.tmdb.org/t/p/original/4nssBcQUBadCTB...	Cinderella has faith her dreams of a better life w...	7.3	2022-02-27
tt0046183	Peter Pan	1953-02-05	77	4000000	87404651	https://image.tmdb.org/t/p/original/fJJOs1iyrhKfZc...	Leaving the safety of their nursery behind, Wendy,...	7.3	2022-07-13
tt0048280	Lady and the Tramp	1955-06-22	76	4000000	36359037	https://image.tmdb.org/t/p/original/wXKeGea4htM7Yz...	Lady, a golden cocker spaniel, meets up with a mon...	7.3	2022-10-18
tt0053285	Sleeping Beauty	1959-02-17	75	6000000	51600000	https://image.tmdb.org/t/p/original/j2mTyUukcLwDle...	A beautiful princess born in a faraway kingdom is ...	7.2	2022-02-27
tt0055254	One Hundred and One Dalmatians	1961-01-25	79	4000000	85000000	https://image.tmdb.org/t/p/original/9klDisS1sVb5Ll...	When a litter of dalmatian puppies are abducted by...	7.3	2022-02-27
tt0057546	The Sword in the Stone	1963-12-25	79	3000000	22182353	https://image.tmdb.org/t/p/original/7lyeeuhGAJSNXY...	Wart is a young boy who aspires to be a knight's s...	7.2	2022-02-27
tt0058331	Mary Poppins	1964-08-27	139	6000000	103082380	https://image.tmdb.org/t/p/original/yO8UyO86uNBq12...	A magical nanny employs music and adventure to hel...	7.8	2022-02-27

Figura 1: Sample

2 Deployment

Docker pull command

```
docker pull redis
```

Server (first time)

```
docker run --network bigdata --name my-redis -d redis
```

If the network bigdata does not exist you should create it before:

```
docker network create bigdata
```

Server (if already exists)

```
docker start my-redis
```

Client shell

```
docker exec -it my-redis redis-cli
```

Quick start

```
ping
```

3 Basic commands

1. Store and then retrieve a string "Aldo Maccione"associated to the key "actor:1".

2. Store in a list ”movies:sw”the movies ”The phantom menace”and ”Attack of the Clones”.
3. Return the length of the list ”movies:sw”.
4. Return (and remove) the element from the head of the list ”movies:sw”.

3.1 Queues and stacks

1. Implement a queue (FIFO) using a list (for example with values 1, 2, 3, 4 and 5) using LPUSH. In particular how to design ”enqueue”and ”dequeue”.
2. Implement a queue using RPUSH.
3. Implement a stack (LIFO) using a list (for example with values 1, 2, 3, 4 and 5) using LPUSH. In particular how to design ”push”and ”pop”.
4. Implement a stack using RPUSH.

4 Data caching

4.1 Definition

Data caching consists in storing data temporarily in a cache, removing it automatically after a given period of time or relying on a replacement policy when the cache is full. Such a technique is used in distributed systems, particularly web applications in order to improve performances. Indeed, on one hand caching enables data to be closer to the application or users, on the other hand it leads to reduce the number of queries to the database/storage system.

4.2 Alternatives

Several alternatives exist to store SQL queries results in Redis, depending on the application's needs and the data structure to be used:

1. Full result caching: Entire result of the SQL query in Redis using a unique key to identify the result. Such an approach allows to retrieve the data directly from the key-value store instead of executing the SQL query every time.
2. Individual result caching: Each individual result using unique keys. For example, Redis hashes can be used to store each row of the query, the key being a unique identifier of the record and the fields corresponding to the table columns.
3. Pagination caching: If the result is large, Redis can be used to cache paginated data. That is to say dividing the results into pages and then storing the pages with a unique key per page. As a consequence, subsets of data can be quickly accessed without considering the whole result.

4.3 Validation

4.3.1 Full result caching / Query caching

Let the query `SELECT * FROM MOVIE WHERE RUNTIME=143` associated to the following records:

```
tt0325980,Pirates of the Caribbean: The Curse of the Black Pearl,2003-07-09,143,140000000,655011224,htt
tt1477834,Aquaman,2018-07-06,143,160000000,1148461807,https://image.tmdb.org/t/p/original/xLPffWMhMj1l5
```

1. Store the previous query as a string in the CSV format.
2. Store the previous query as a string in the JSON format.

4.3.2 Individual result caching / Tuple caching

Let the 83 minute movie 'Snow White and the Seven Dwarfs', released on 1937-12-21, associated with the id tt0029583. Its budget was 1488423$, its revenue is 184925486 $ and its rating is 7.6. Its cover can be seen at https://image.tmdb.org/t/p/original/1P9eGGlT7eV7kAAvvSh9jfygr1C.jpg.

1. Store 'Snow White and the Seven Dwarfs' as a CSV string. Retrieve the size of this entry.
2. Retrieve the size of this CSV like entry.
3. Store 'Snow White and the Seven Dwarfs' as a hash.
4. Retrieve the size of this hash entry.

4.4 Web application acceleration

The objective of this activity is to provide a simple web application with acceleration using data caching.

1. Propose a simple baseline web architecture with for example a Relational Data Base Management System (DBMS).
2. Integrate a caching layer is the previous architecture.
3. Define an experimental protocol to compare the two solutions.

5 References

- **Lecture**: https://perso.univ-rennes1.fr/laurent.dorazio/data/teachings/big_data/big_data_redis.pdf
- **Redis documentation**: https://redis.io/docs/